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USGS Earthquake Data Analysis Engineering Project Report

Introduction

Earthquakes can have devastating effects on buildings and infrastructure, potentially endangering human lives. Understanding earthquake magnitude, depth, frequency, and location is important from an engineering perspective, particularly when designing structures that need to withstand seismic activity. Both stronger earthquakes and repeated smaller ones can take their toll on infrastructure over time. For this project, I chose to analyze earthquake data from the United States Geological Survey (USGS) using Python. The goal was to examine the relationship between earthquake magnitude and depth, identify patterns in earthquake frequency and location, and determine which events may be most important from an engineering perspective.

Data

I chose to export and utilize a dataset from the USGS Earthquake Catalog covering a 30-day period, from October 9 to November 8, 2026. The dataset contains 759 recorded earthquakes with magnitudes of 4.5 or greater. The main variables I chose to focus on were time, latitude, longitude, depth, magnitude, place, and event type. These variables allowed me to examine when and where earthquakes occurred, how strong they were, and how deep below the surface they originated. The dataset was imported into Google Colab as a CSV file using Python and Pandas. I checked the data for missing values and found that none of the seven selected columns contained missing entries.

Methods

Python, Pandas, and Matplotlib were used to process, analyze, and visualize the earthquake data. I started by finding the total number of earthquakes, average magnitude, largest and smallest magnitudes, and average, maximum, and minimum depths. I then created two functions to classify the earthquakes one for magnitude and one for depth. Earthquakes were categorized by magnitudes according to how they normally are: below 5.0 were classified as light, 5.0 to below 6.0 as moderate, 6.0 to below 7.0 as strong, 7.0 to below 8.0 as major, and 8.0 or greater as great. The earthquakes occurring at depths below 70 km were classified as shallow, those between 70 and 300 km as intermediate, and those deeper than 300 km as deep.They were then filtered by another function to identify if their magnitudes were 5.0 and above.

Finally, I created four graphs using Matplotlib: a histogram of earthquake magnitudes, a scatterplot comparing magnitude and depth, a bar chart of magnitude categories, and a location-based scatterplot using latitude and longitude.

Results

The dataset contained 759 earthquakes, with an average magnitude of approximately 4.85 and an average depth of 55.88 km, with the largest recorded magnitude was 7.7, while the smallest was 4.5. The deepest earthquake occurred at 642.624 km, while the shallowest was recorded at 1 km. The majority of earthquakes were relatively weak. Of the 759 events, 549 had magnitudes below 5.0, while 187 were moderate, 21 were strong, and only 2 were classified as major. No great earthquakes were recorded.

Most earthquakes were also shallow, with 608 occurring at depths below 70 km. There were only 116 intermediate-depth earthquakes and 35 deep earthquakes.

The magnitude histogram showed that earthquake frequency decreased significantly as magnitude increased. The magnitude versus depth scatterplot showed that most earthquakes were relatively shallow, although earthquakes of similar magnitudes occurred at different depths.

The longitude v latitude plot showed that earthquakes were concentrated in certain geographic regions rather than evenly distributed across the world.

Engineering Interpretation

Earthquakes can have significant impacts on buildings and infrastructure, making seismic activity an important consideration in structural and civil engineering. Stronger earthquakes have the potential to cause serious structural damage, including cracks in foundations, failure of load-bearing components, and even the collapse of buildings. This can endanger human lives and disrupt essential services, particularly when roads, bridges, power systems, and water infrastructure are affected.

Magnitude and depth are both important factors when considering the potential damage caused by earthquakes. Stronger earthquakes release more energy, which can result in more intense ground shaking over larger areas. Earthquakes occurring closer to the surface can also have a more direct impact on nearby structures than deeper earthquakes otherwise would. However, the extent of the damage also depends on factors such as distance from the epicenter, local soil conditions, and the structural integrity of buildings. Repeated earthquakes can also take their toll on infrastructure over time. Even when individual earthquakes do not cause major damage, repeated shaking can contribute to the deterioration of structures, particularly those that are older or were not designed to withstand seismic activity. This makes proper structural design, regular inspections, and maintenance important in earthquake-prone regions.

From an engineering perspective, analyzing earthquake frequency, magnitude, depth, and location can help identify areas that may be more vulnerable to seismic activity. This information can be used to inform building codes, improve structural designs, and prioritize infrastructure improvements. Understanding these patterns is particularly important when designing structures intended to remain safe and functional during and after an earthquake.

Limitations and Conclusion

One major limitation of my analysis is the dataset itself. I chose to examine only one month of earthquakes with magnitudes of 4.5 or greater. This is an extremely limited dataset in the grand scheme of things, as it does not provide a full picture of longer-term earthquake patterns or the impact of weaker earthquakes. If I were to perform another study, I would choose to analyze a longer period and include weaker earthquakes. I would also want to pay closer attention to the actual damage caused by these events, particularly to buildings and infrastructure. Additional data on structural damage would help determine how earthquakes affect the built environment.

Overall, the results showed that most recorded earthquakes these past thirty days were relatively weak and shallow, while only a small number were classified as major. Although this analysis is limited, it demonstrates how earthquake data can be used to better understand seismic activity and its importance to structural and infrastructure engineering.

Data Source

United States Geological Survey (USGS), Earthquake Catalog.
https://earthquake.usgs.gov/earthquakes/search/